

Figure 17: Cosine similarity of target-token hidden states across layers in Mistral-7B-v0.3. From mid-layers onward, FAs and AHs diverge sharply as subject information propagates, while UHs remain more clustered, confirming weaker subject-dependent updates.

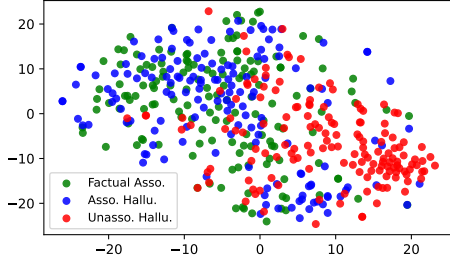


Figure 18: t-SNE visualization of subject tokens' representations at layer 11 of LLaMA-3-8B.

$$\text{Ratio}_{\text{AH/FA}}^{(\ell)} = \frac{\bar{N}_{\text{AH}}^{(\ell)}}{\bar{N}_{\text{FA}}^{(\ell)}} \quad (4)$$

$$\text{Ratio}_{\text{UH/FA}}^{(\ell)} = \frac{\bar{N}_{\text{UH}}^{(\ell)}}{\bar{N}_{\text{FA}}^{(\ell)}} \quad (5)$$

C Parallel Experiments on Mistral

This section is for documenting parallel experiments conducted on the Mistral-7B-v0.3 model under the same settings as described in the main text (Figures 12–21).

The results from Mistral exhibit similar patterns to those observed in LLaMA, as described before. Specifically, we find consistent patterns in the model's internal computations, hidden-state behaviors, and the performance of hallucination detection and refusal tuning experiments.

D Further Analysis on Internal States

D.1 Analysis of Parametric Subspace Overlap for Subject Representation

We investigate why early layers encode subject representations differently across knowledge types by examining how inputs interact with the parametric

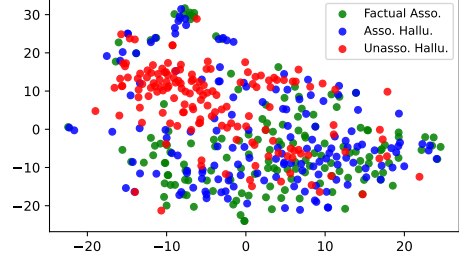


Figure 19: t-SNE visualization of subject tokens' representations at layer 11 of Mistral-7B-v0.3.

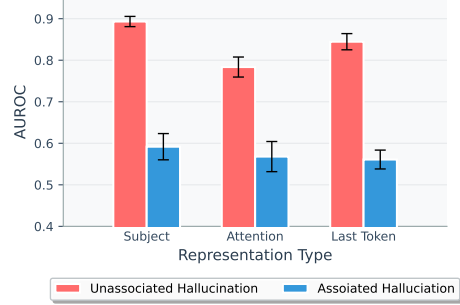


Figure 20: Hallucination detection performance on the Full setting (Mistral-7B-v0.3).

knowledge stored in MLP modules. Inspired by Kang et al. (2024), the output norm of an MLP layer depends on how well its input aligns with the subspace spanned by the weight matrix: *poorly aligned inputs yield smaller output norms*.

For each MLP layer ℓ , we analyze the down-projection weight matrix W_{down}^{ℓ} and its input x^{ℓ} . Given the input x_s^{ℓ} corresponding to the subject tokens, we compute its overlap ratio with the top singular subspace V_{top} of W_{down}^{ℓ} :

$$r(x_s^{\ell}) = \frac{\|x_s^{\ell \top} V_{\text{top}} V_{\text{top}}^{\top}\|^2}{\|x_s^{\ell}\|^2}. \quad (6)$$

A higher overlap ratio $r(x_s^{\ell})$ indicates stronger alignment to the subspace spanned by W_{down}^{ℓ} , leading to larger output norms.

To highlight relative deviations from the factual baseline (FA), we report the relative ratios between AH/FA and UH/FA. Focusing on the layer with the largest UH norm shift, Figure 22 shows that UHs have significantly lower $r(x_s^{\ell})$ than AHs in both LLaMA and Mistral. This reveals that early-layer parametric weights are more aligned with FA and AH subject representations than with UH subjects, producing higher norms for the former ones. These results also suggest that the model has sufficiently learned representations for FA and AH subjects during pretraining but not for UH subjects.

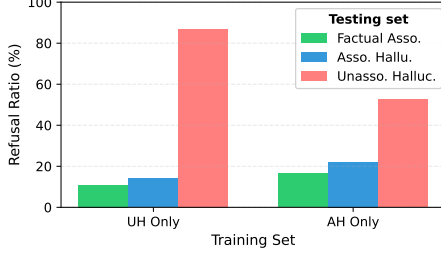


Figure 21: Refusal tuning performance across three types of samples (Mistral-7B-v0.3).

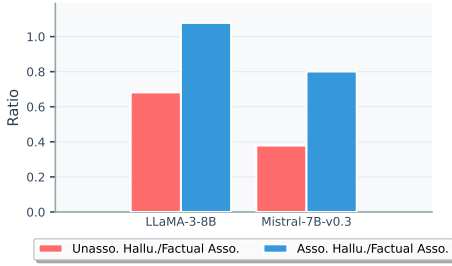


Figure 22: Comparison of subspace overlap ratios.

D.2 Quantitative Analysis of Cluster Separability

To quantitatively validate the visual separability observed in the t-SNE visualizations (Figures 7 and 14), we compute two cluster separability metrics on the hidden representations: the Silhouette coefficient and the Davies-Bouldin Index (DBI). The Silhouette coefficient ranges from -1 to 1 , where higher values indicate better-defined clusters. The DBI measures the average similarity ratio of each cluster with its most similar cluster, where lower values indicate better separation.

The results shown in Table 4 quantitatively confirm our observations from the t-SNE visualizations. When comparing UHs and FAs, both models exhibit high Silhouette coefficients (0.33–0.40) and low DBI values (0.93–1.13), indicating clear cluster separation between UHs and FAs. In contrast, when comparing AHs and FAs, the Silhouette coefficients are near zero (0.0022 and -0.0011) and DBI values are substantially higher (14.22 and 25.25), indicating strong overlap between AHs and FAs. These metrics support our claim that FAs and UHs form well-separated clusters in the representation space, while AHs strongly overlap with FAs, consistent with our hypothesis that AHs rely on similar knowledge-recall processes as FAs.

	UH Only		AH Only	
Model	Sil. $\uparrow$	DBI $\downarrow$	Sil. $\uparrow$	DBI $\downarrow$
LLaMA	0.397	0.932	0.002	14.22
Mistral	0.327	1.134	-0.001	25.25

Table 4: Quantitative cluster separability metrics. Higher Silhouette (Sil., $\uparrow$) and lower Davies-Bouldin Index (DBI, $\downarrow$) indicate better separation.

Model	Method	UH Only	AH Only
Mistral	Prob.	.894 (.889–.899)	.457 (.453–.461)
	Subj. Pop.	.812 (.787–.839)	.570 (.551–.590)
	Subject	.893 (.878–.909)	.592 (.552–.631)
	Attention	.872 (.864–.881)	.584 (.500–.668)
	Last Token	.923 (.913–.934)	.629 (.610–.648)
LLaMA	Prob.	.860 (.853–.867)	.488 (.480–.495)
	Subj. Pop.	.869 (.856–.883)	.479 (.467–.491)
	Subject	.913 (.897–.930)	.648 (.626–.669)
	Attention	.920 (.889–.950)	.579 (.525–.634)
	Last Token	.935 (.923–.947)	.691 (.660–.722)

Table 5: Hallucination detection performance with 95% confidence intervals across five random seeds.

D.3 More Visualization on Hidden States

In this section, we provide t-SNE visualization of subject tokens’ hidden states in Figure 18 and Figure 19.

Compared to the last-token representations, the t-SNE visualization of subject-token hidden states shows that unassociated hallucinations (UHs) are moderately separated from factual and associated samples, they exhibit marginally lower separability than last-token representations. This observation aligns with the results in §5, where the hallucination detection performance using last-token hidden states outperforms that based on subject-token representations.

E Further Analysis on Hallucination Detection

E.1 Statistical Significance Analysis

We report the hallucination detection performance (§5) with 95% confidence intervals across five random seeds in Table 5. Across all settings relevant to our claims, AUROC on UHs is significantly higher than on AHs (all $p < 0.01$), supporting our finding that current detection methods are substantially more effective for UHs than AH.

E.2 Semantic Consistency-Based Detection

We sample 200 instances per category and generate 5 responses per prompt with temperature = 0.7. We

Model	UH Only	AH Only
LLaMA	0.8374	0.6183
Mistral	0.8437	0.6159

Table 6: Semantic consistency-based hallucination detection performance.

Threshold	UH Only		AH Only	
	Sensitivity	Specificity	Sensitivity	Specificity
0.1	0.900	0.833	0.697	0.583
0.3	0.879	0.888	0.644	0.633
0.5	0.867	0.925	0.601	0.679
0.7	0.867	0.942	0.534	0.725
0.9	0.838	0.963	0.433	0.771

Table 7: Sensitivity-specificity trade-offs at different thresholds.

compute a consistency score for each sample and evaluate hallucination detection performance using these scores, measured by AUROC. Table 6 shows the results.

The results show that the performance of this consistency-based method aligns with our findings for both the black-box and white-box methods discussed in §5: it can effectively distinguish UHs from FAs (AUROC ≈ 0.84) but struggles to separate AHs from FAs (AUROC ≈ 0.62). This behavioral evidence supports our mechanistic analysis: since AHs rely on similar knowledge-recall processes as FAs, they tend to produce similarly consistent outputs across runs, whereas UHs exhibit higher variance due to weaker reliance on subject representations.

E.3 Sensitivity-Specificity Trade-offs

To further examine the detection performance beyond AUROC, we analyze the trade-offs between sensitivity and specificity at various classification thresholds. We report results for detectors trained on LLaMA’s last-token representations. Tables 7 present the results.

For UH Only, we observe that detection performance is remarkably robust across the threshold range. Even as the threshold increases from 0.1 to 0.9, the detector maintains a high balance of metrics, with sensitivity remaining above 0.83 and specificity exceeding 0.96 at the highest setting. This stability indicates a clear and consistent separation in the representation space. In contrast, for AH Only, the performance is highly sensitive to the specific threshold. As the threshold moves from 0.1 to 0.9, sensitivity drops significantly from 0.697 to

0.433, while specificity only improves from 0.583 to 0.771. This trade-off suggests that establishing a stable decision boundary between AHs and FAs is challenging, confirming that their internal representations strongly overlap.